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Executive Summary:

The purpose of this report is to analyse why the telecommunication company is experiencing churn. Three statistical analyses were conducted: descriptive, diagnostic, and predictive. The findings presented that the main drivers of churn were lower tenure, higher monthly charges, and lack of technical support. The analysis found that if customers don't perceive value in the price charged, they are very likely to leave altogether. All the tools used align with the same conclusion, even logistic regression, as the overall accuracy is 79.5%

Introduction: -

Customer churn is a major concern in highly competitive service industries, such as telecommunications. (Ahmad, Jafar and Aljoumaa, 2019) Customer churn means that customers are leaving the company. Churn could occur due to many factors, such as intense competition or service offerings that are not up to the mark. If churn rate profitability can be significantly affected, as obtaining new customers is expensive. Retaining an existing customer is much cheaper than obtaining a new one. (King *et al.*, 2016)

As a result, the telecommunications company must identify the factors driving churn and the retention strategies it can implement. The telecommunication company has been offering a range of services, and retention remains a common problem. One reason could be a mismatch between customer expectations and the value delivered. New and price-sensitive customers may not perceive value in the services being offered, leading to high churn.

Therefore, the primary aim of this study is to identify the key factors influencing customer churn by analysing behavioral, financial, service, and contractual variables. After identifying the factors driving churn, actionable recommendations will be provided.

Data Summary

The dataset includes one dependent variable, churn, and multiple independent variables, such as tenure, internet type, and streaming services. According to Figure 2, there were 7043 customers and 21 variables. There are only three continuous variables (Tenure, Monthly Charges, and Total Charges); the rest are categorical. Key variables that were used in this research were tenure, monthly charges, total charges, contract type, technical support, online security, types of internet service, and paperless billing.

Data Cleaning: -

The data was checked before any statistical analysis was performed. There were many blank sections in the monthly charges, which were then replaced with the value 0. In the Excel file, churn was recoded (0 = No, 1 = Yes), and a new column, churn binary, was created to run logistic regression. The document was scanned for outliers too but none were found.

Descriptive statistics: -

Figure 2 summarizes descriptive statistics for three continuous variables: tenure, monthly charges, and total charges.

MonthlyCharges		*TotalCharges*		*tenure*	
Mean	64.76169246	Mean	2279.734304	Mean	32.37114866
Standard Error	0.358545291	Standard Error	27.01054208	Standard Error	0.292644483
Median	70.35	Median	1394.55	Median	29
Mode	20.05	Mode	20.2	Mode	1
Standard Deviation	30.0900471	Standard Deviation	2266.79447	Standard Deviation	24.55948102
Sample Variance	905.4109343	Sample Variance	5138357.168	Sample Variance	603.1681081
Kurtosis	-1.257259695	Kurtosis	-0.228579807	Kurtosis	-1.387371636
Skewness	-0.220524434	Skewness	0.963234655	Skewness	0.23953975
Range	100.5	Range	8684.8	Range	72
Minimum	18.25	Minimum	0	Minimum	0
Maximum	118.75	Maximum	8684.8	Maximum	72
Sum	456116.6	Sum	16056168.7	Sum	227990
Count	7043	Count	7043	Count	7043

Figure 2: Descriptive statistics

The average customer tenure is 32.37, indicating a significantly long relationship with the company. The mean monthly charge is £64.76, with a wide range of £18.25 to

£118.75, suggesting significant variation in pricing among customers. Total charges show the highest variability (SD = 2266.79), reflecting differences in how long customers have been with the company.

Variable	Category	Count	Percentage
Churn	No	5174	73.46%
Churn	Yes	1869	26.54%
	Grand Total	7043	100.00%

Figure 2: Customer Churn Data

Table 2 shows the churn distribution. The total number of customers is 7043; most customers did not churn (73.46%), and only 26.54% churned. This indicates a class imbalance in the dataset, with non-churners significantly outnumbering churners.

Figure three shows the gender distribution of customers. The image clearly shows that males (3,555) and females (3,488) are evenly split. This indicates that gender is not a significant factor in this analysis.

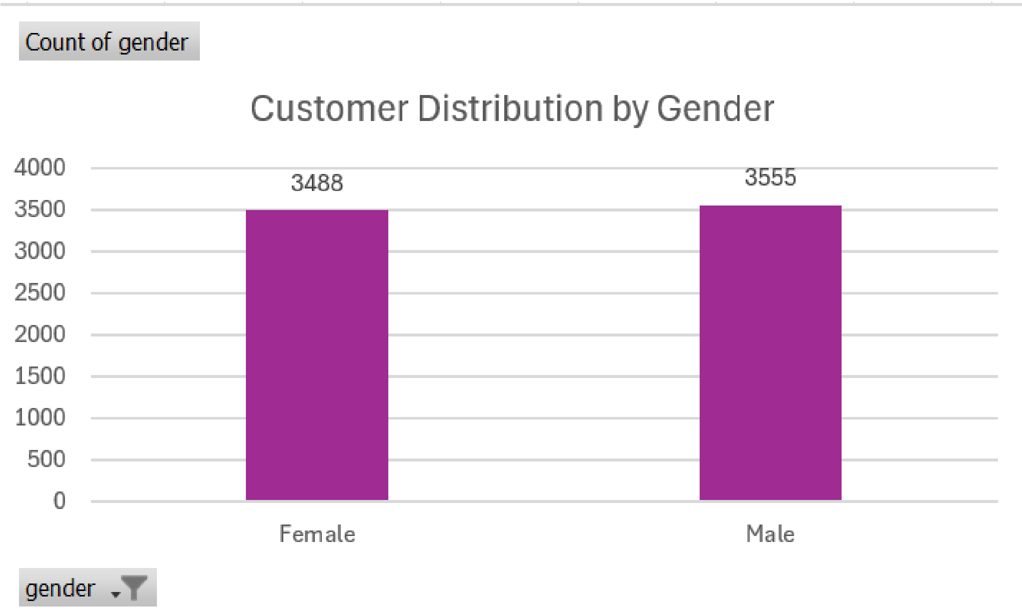


Figure 3: Gender distribution.

Methodology: -

This report analyses what factors are leading to customer churn in the telecommunications industry. To determine the factors leading to customer departures, three statistical processes were conducted: descriptive statistics, diagnostics, and predictive analytics. Descriptives were used to identify trends and patterns in the data, with visuals created in two software programs: Tableau and Excel. In this report, many visuals will be found, such as histograms, bar charts, and box-and-whisker plots.

Diagnostics were used to assess whether these variables were significantly related. For example, chi-square is used four times to check associations between the independent variable and the dependent variable, such as whether internet service has a positive or a negative association with churn. Other variables that were used in the chi-square test were online security, contracts, and technical support. Two independent t-tests were conducted to examine the difference in tenure and monthly charges between those who stayed and those who churned. Each test was accompanied by a null hypothesis (H0) and an alternative hypothesis (H1) to provide a structured framework for evaluating statistical significance. The last diagnostic was done by correlation to examine the relationship between continuous variables. The diagnostics were done by using the software SPSS.

For predictive analysis, SPSS was used to perform logistic regression. The model was used in predicting how likely customers are to churn when all significant variables are brought together. This method makes sure that the methodology connects to the business problem by moving from exploration to confirmation to prediction.

Results: -

Descriptive:

Customer Behaviour and Churn:

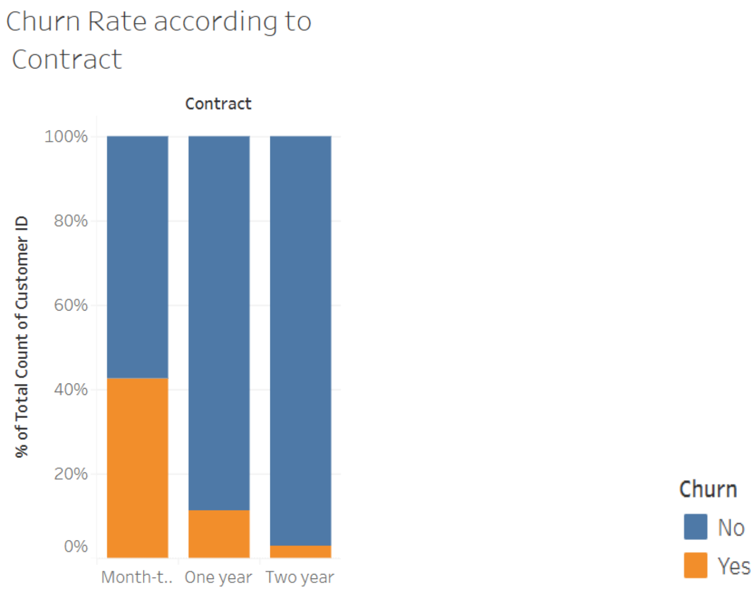


Figure 4: Contract vs Churn

The bar chart above shows a strong relationship between customer contract type and churn. The analysis indicates that customers on long-term contracts are less likely to churn than those on monthly contracts. However, customers on a monthly contract have a 40% churn rate. This indicates that long-term commitments are far less likely to switch providers; thus, it amplifies that longer contracts act as a retention strategy.

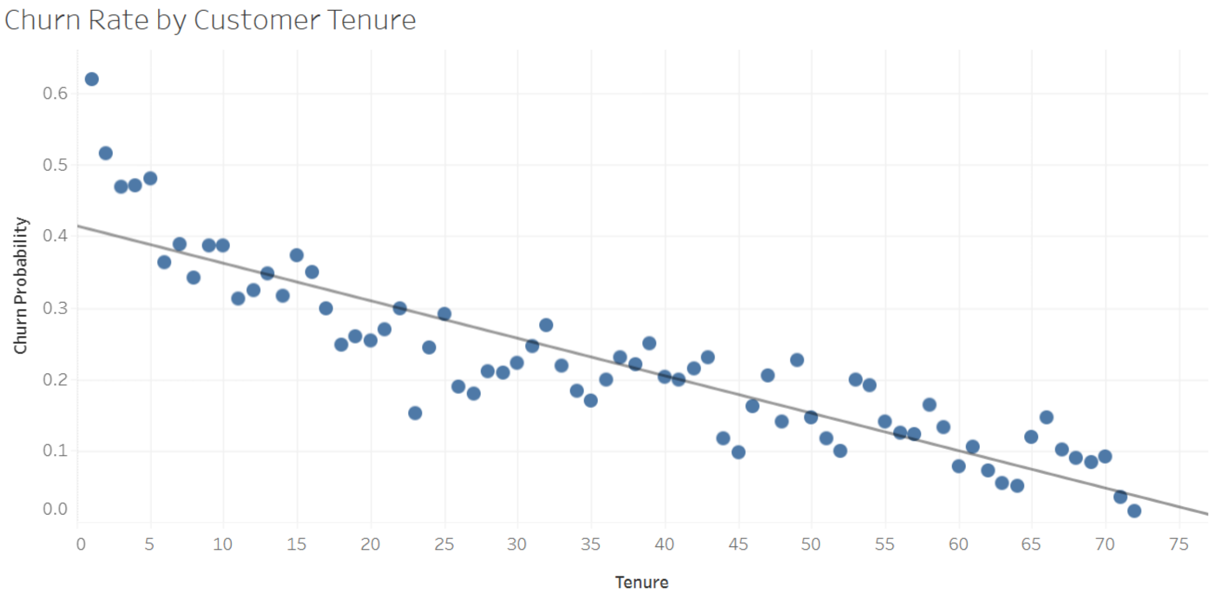


Figure 5: Tenure vs Churn

As the figure above shows, the churn rate is higher among customers with shorter tenure. Customers who have just joined the telecom company are more likely to leave. It can be observed that customers are more susceptible in the early stages of their relationship with a company.

Service factors influencing Churn: -

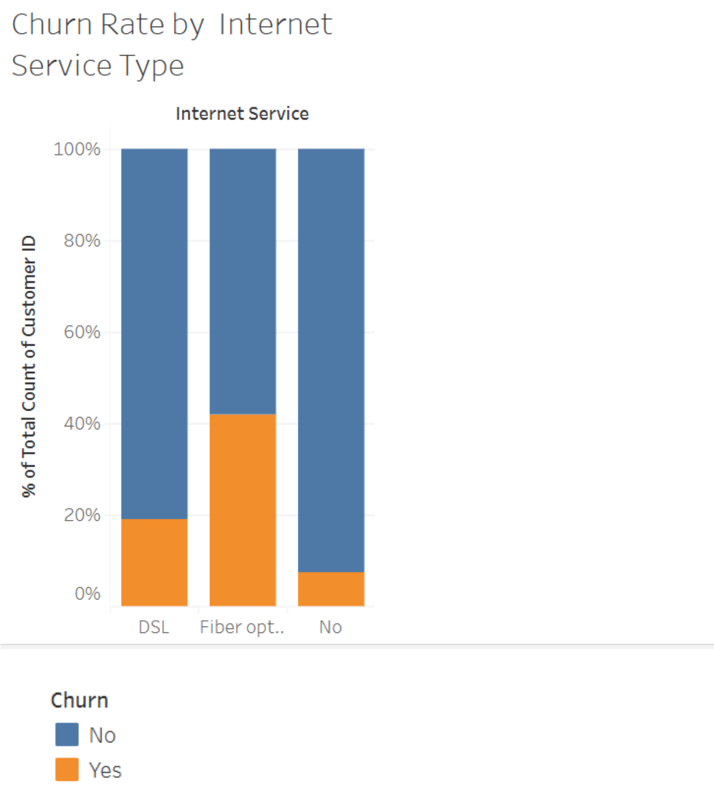


Figure 6: Internet Service vs Churn

The clustered bar chart shows that churn varies across DSL, fiber optics, and no internet at all. Fiber optics has the highest churn rate, at about 41.89%, while DSL has 18.96%, which is significantly lower. On the other hand, there are some customers who use the telecom company for a different reason altogether, as the churn rate is only 7.40%

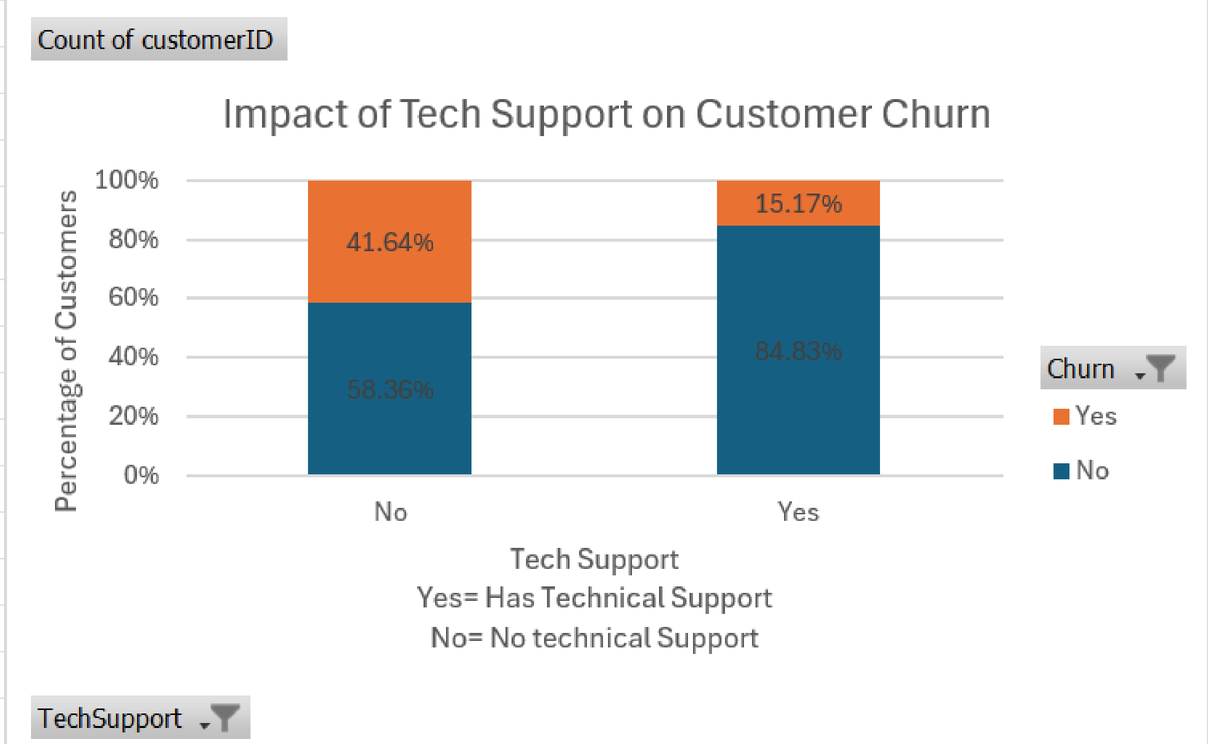


Figure 7: Tech Support vs Churn

The chart above shows that churn rates vary significantly depending on the company providing technical support. Customers who do not receive technical support have a churn rate of 41.64%, highlighting the value of this additional service. In contrast, those with technical support experience a much lower churn rate of 15.17%. This analysis indicates that customers greatly appreciate the availability of technical support.

Financial Factors Influencing Churn

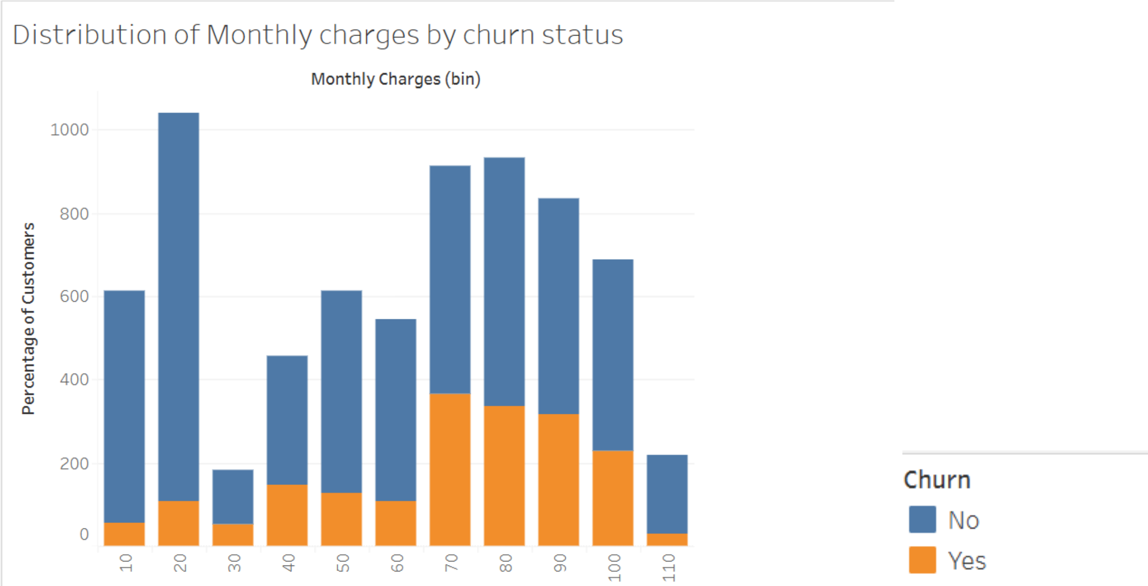


Figure 8: Monthly charges vs Churn

Different price ranges have different effects on churn; the histogram shows that churned and non-churned customers are present across all price ranges. The chart indicates that a large proportion of customers fall within the mid- to higher-price range, with both groups represented. It can be observed that customers are not fixed at one price level; as shown in the graph, they are distributed. However, the relationship is not fully consistent across the entire price range. To further examine this relationship, a box plot is presented below:

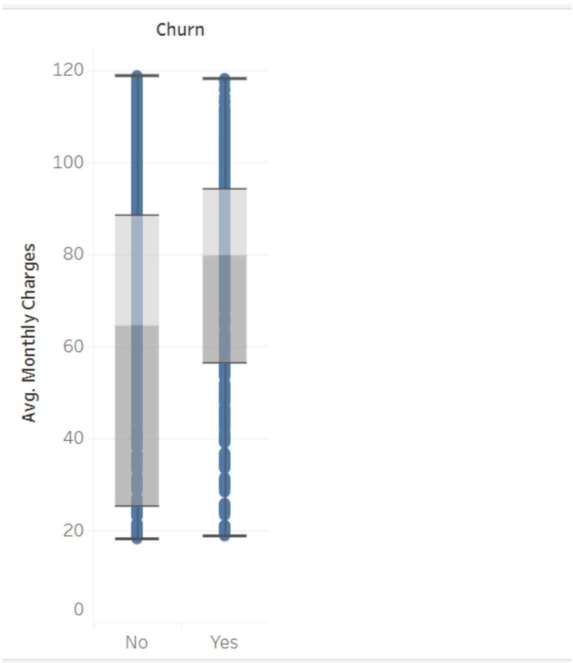
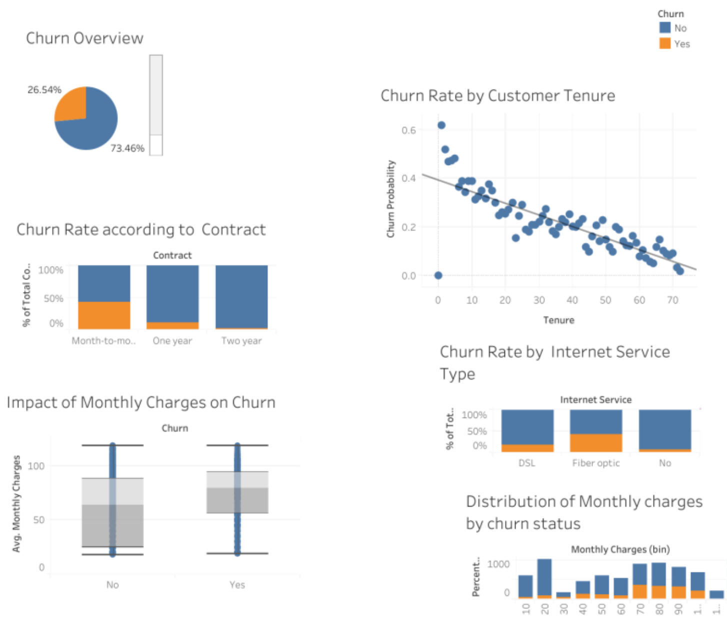


Figure 9: Whiskers and Plot.

The box plot clearly compares churned and non-churned customers. From the box chart above, it is evident that new customers often pay a higher monthly charge than existing customers. The median monthly charge is higher, indicating a positive relationship between pricing and churn.

Tableau:



Diagnostics:

- T-Test:

Hypothesis:-

H0: There is no significant difference in tenure between churned and non-churned customers.

H1: There is a significant difference in tenure between churned and non-churned customers.

T-Test

Group Statistics

	Churn binary	N	Mean	Std. Deviation	Std. Error Mean
tenure	0	5174	37.57	24.114	.335
	1	1869	17.98	19.531	.452

Table:-

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means							
						Significance				95% Confidence Interval of the Difference	
		F	Sig.	t	df	One-Sided p	Two-Sided p	Mean Difference	Std. Error Difference	Lower	Upper
tenure	Equal variances assumed	336.653	<.001	31.580	7041	<.001	<.001	19.591	.620	18.375	20.807
	Equal variances not assumed			34.824	4048.288	<.001	<.001	19.591	.563	18.488	20.694

Figure:

Interpretation:-

Levene's test for equality of variance was significant (F=336.653, p<.001), showing us that equal variances could not be assumed. Hence, the equal variance not assumed row was used. The test revealed a significant difference in tenure between churned and non-churned customers (t = 34.824, df = 4048.288, p < .001). Customers who left had a significantly lower mean tenure of 17.98, compared to those who stayed with the company, who had a mean tenure of 37.57. From here, we can conclude that newer customers are more likely to churn. Therefore, H0 is rejected.

- **Second Independent t-test:-**

  Hypothesis

$H0_2$: There is no significant difference in monthly charges between churned and non-churned customers.

$H1_2$: There is a significant difference in monthly charges between churned and non-churned customers.

T-Test

Group Statistics

	Churn binary	N	Mean	Std. Deviation	Std. Error Mean
MonthlyCharges	0	5174	61.2651	31.09265	.43226
	1	1869	74.4413	24.66605	.57055

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means							
						Significance		Mean Difference	Std. Error Difference	95% Confidence Interval of the Difference	
		F	Sig.	t	df	One-Sided p	Two-Sided p			Lower	Upper
MonthlyCharges	Equal variances assumed	358.128	<.001	-16.537	7041	<.001	<.001	-13.17621	.79678	-14.73815	-11.61427
	Equal variances not assumed			-18.408	4135.795	<.001	<.001	-13.17621	.71581	-14.57957	-11.77284

Interpretation: -

Levene's test for equality of variances was significant ($F = 358.128$, $p < .001$), indicating that equal variances could not be assumed. Therefore, the equal variances not assumed row was used. The independent-samples t-test indicates a statistically significant difference in monthly charges between churned and non-churned customers ($t = -18.408$, $df = 4135.795$, $p < .001$). Customers who left the telecom company have a higher average monthly charge (74.44) than those who stayed (61.26). This relationship is further supported by the descriptive chart box and plot. As a result, the null hypothesis is rejected.

- **Chi-square:-**

Hypothesis:

$H0_3$: There is no association between tech support and customer churn.

$H1_3$: There is a significant association between tech support and customer churn.

Churn binary * TechSupport Crosstabulation

			TechSupport No	TechSupport No internet service	TechSupport Yes	Total
Churn binary	0	Count	2027	1413	1734	5174
		% within Churn binary	39.2%	27.3%	33.5%	100.0%
		% within TechSupport	58.4%	92.6%	84.8%	73.5%
	1	Count	1446	113	310	1869
		% within Churn binary	77.4%	6.0%	16.6%	100.0%
		% within TechSupport	41.6%	7.4%	15.2%	26.5%
Total		Count	3473	1526	2044	7043
		% within Churn binary	49.3%	21.7%	29.0%	100.0%
		% within TechSupport	100.0%	100.0%	100.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	828.197[a]	2	<.001
Likelihood Ratio	887.714	2	<.001
N of Valid Cases	7043		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 404.95.

Interpretation:

The chi-square test shows a significant association between technical support and customer churn ($\chi2$ = 828.197, $p < .001$). Customers who don't have access to technical support are more likely to churn (77.4%), while only 16.6% have access to it. This analysis shows that a lack of technical support is strongly associated with higher churn rates. Hence, the null hypothesis is rejected.

- **Second Chi-Square**

  Hypothesis:

$H0_4$: There is no association between contract type and customer churn.

$H1_4$: There is a significant association between contract type and customer churn.

Churn binary * Contract Crosstabulation

			Contract			
			Month-to-month	One year	Two year	Total
Churn binary	0	Count	2220	1307	1647	5174
		% within Churn binary	42.9%	25.3%	31.8%	100.0%
		% within Contract	57.3%	88.7%	97.2%	73.5%
	1	Count	1655	166	48	1869
		% within Churn binary	88.6%	8.9%	2.6%	100.0%
		% within Contract	42.7%	11.3%	2.8%	26.5%
Total		Count	3875	1473	1695	7043
		% within Churn binary	55.0%	20.9%	24.1%	100.0%
		% within Contract	100.0%	100.0%	100.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	1184.597[a]	2	<.001
Likelihood Ratio	1386.810	2	<.001
N of Valid Cases	7043		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 390.89.

Interpretation:

According to the test, there is a significant association between contract type and customer churn ($\chi 2$ = 1184.597, p <.001). Customers on month-to-month contracts have a much higher churn rate (88.6%) than those on one-year (8.9%) or two-year contracts (2.6%). Hence, the null hypothesis is rejected.

- **Third chi-square:**

  Hypothesis:

$H0_5$: There is no association between internet service type and customer churn.

$H1_5$: There is a significant association between internet service type and customer churn.

Churn binary * InternetService Crosstabulation

			InternetService			
			DSL	Fiber optic	No	Total
Churn binary	0	Count	1962	1799	1413	5174
		% within Churn binary	37.9%	34.8%	27.3%	100.0%
		% within InternetService	81.0%	58.1%	92.6%	73.5%
	1	Count	459	1297	113	1869
		% within Churn binary	24.6%	69.4%	6.0%	100.0%
		% within InternetService	19.0%	41.9%	7.4%	26.5%
Total		Count	2421	3096	1526	7043
		% within Churn binary	34.4%	44.0%	21.7%	100.0%
		% within InternetService	100.0%	100.0%	100.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	732.310[a]	2	<.001
Likelihood Ratio	782.818	2	<.001
N of Valid Cases	7043		

a. 0 cells (0.0%) have expected count less than 5. The minimum expected count is 404.95.

Interpretation: -

There is a significant association between internet service and churn ($\chi^2 = 732.310$, df = 2, $p < .001$). Hence, the null hypothesis is rejected.

- **Fourth Chi-Square**

Hypothesis:

H0: There is no association between online security and customer churn.

H1: There is a significant association between online security and customer churn.

Churn binary * OnlineSecurity Crosstabulation

			OnlineSecurity No	OnlineSecurity No internet service	OnlineSecurity Yes	Total
Churn binary	0	Count	2037	1413	1724	5174
		% within Churn binary	39.4%	27.3%	33.3%	100.0%
		% within OnlineSecurity	58.2%	92.6%	85.4%	73.5%
	1	Count	1461	113	295	1869
		% within Churn binary	78.2%	6.0%	15.8%	100.0%
		% within OnlineSecurity	41.8%	7.4%	14.6%	26.5%
Total		Count	3498	1526	2019	7043
		% within Churn binary	49.7%	21.7%	28.7%	100.0%
		% within OnlineSecurity	100.0%	100.0%	100.0%	100.0%

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	849.999[a]	2	<.001
Likelihood Ratio	911.044	2	<.001
N of Valid Cases	7043		

Interpretation: -

There is a positive, strong association between online security and churn (χ^2 = 849.999, df = 2, $p < .001$). Customers with online security had a lower churn rate (14.6%) than those without (41.8%). Hence, those who obtain online security are less likely to churn compared to those who don't. Therefore, the null hypothesis is rejected.

Correlation: -

Correlations

		tenure	MonthlyCharges	TotalCharges
tenure	Pearson Correlation	1	.248**	.826**
	Sig. (2-tailed)		<.001	<.001
	N	7043	7043	7032
MonthlyCharges	Pearson Correlation	.248**	1	.651**
	Sig. (2-tailed)	<.001		<.001
	N	7043	7043	7032
TotalCharges	Pearson Correlation	.826**	.651**	1
	Sig. (2-tailed)	<.001	<.001	
	N	7032	7032	7032

**. Correlation is significant at the 0.01 level (2-tailed).

The correlation analysis shows a significant relationship between all the variables tested: “tenure”,” MonthlyCharges”, and “TotalCharges”. Tenure has a strong positive correlation with total charges (r=0.826) and a weak positive correlation with monthly charges (r=0.248). Monthly and Total charges also have a strong positive correlation (0.625). All three are statistically significant.

Predictive analysis:

Logistic regression was applied because the dependent variable, which is churn, is binary. This method allows us to estimate the probability of customer churn based on predictive variables. The model incorporates behavioral factors (tenure, paperless billing), financial factors (monthly charges), service-related factors (tech support, online security, and internet service), and contractual factors (contract type) to capture the key drivers of churn.

DV:- Churn (Binary)

IVs:- Tenure, Monthly Charges, Tech Support, contract, paperless billing, Internet Type, and Online Security

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	5961.449[a]	.267	.390

a. Estimation terminated at iteration number 6 because parameter estimates changed by less than .001.

Classification Table[a]

Observed			Predicted: Churn binary 0	Predicted: Churn binary 1	Percentage Correct
Step 1	Churn binary	0	4618	556	89.3
		1	888	981	52.5
	Overall Percentage				79.5

a. The cut value is .500

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 1[a]	tenure	-.032	.002	230.419	1	<.001	.969
	MonthlyCharges	.012	.003	13.413	1	<.001	1.012
	TechSupport			48.729	2	<.001	
	TechSupport(1)	-.960	.260	13.602	1	<.001	.383
	TechSupport(2)	-.450	.087	26.638	1	<.001	.638
	Contract			96.684	2	<.001	
	Contract(1)	-.745	.105	50.722	1	<.001	.475
	Contract(2)	-1.467	.172	72.623	1	<.001	.231
	PaperlessBilling(1)	-.411	.073	31.691	1	<.001	.663
	OnlineSecurity			35.617	1	<.001	
	OnlineSecurity(1)	.505	.085	35.617	1	<.001	1.657
	InternetService			17.515	1	<.001	
	InternetService(1)	-.582	.139	17.515	1	<.001	.559
	Constant	-.402	.297	1.827	1	.176	.669

a. Variable(s) entered on step 1: tenure, MonthlyCharges, TechSupport, Contract, PaperlessBilling, OnlineSecurity, InternetService.

Logistics Equation:-

Logit(Churn) = -0.402 – 0.032(Tenure) + 0.012(MonthlyCharges) – 0.960(TechSupport1) – 0.450(TechSupport2) – 0.745(Contract1) – 1.467(Contract2) – 0.411(PaperlessBilling) + 0.505(OnlineSecurity1) – 0.582(InternetService1)

Interpretation:-

The model demonstrates a reasonably strong predictive performance with an accuracy of 79.5%. However, it is more effective at predicting non-churns (89.3%) than predicting churners (52.5%), suggesting that while the model identifies stable customers well than those who are at the risk of leaving.

Interpretation and Implication:

After running all the types of tests, descriptives, diagnostics, and predictive, it can be concluded that people who churn and who stay have some common characteristics. The descriptive analysis shows that churn is higher among those with shorter tenures and higher monthly charges, indicating early-stage vulnerability and potential dissatisfaction with pricing that does not meet perceived value.

The independent T-test revealed a significant difference in tenure between customers who churned and those who did not ($t = 34.824$, $p<.001$). This shows us that customers who churn have lower tenure. In layman's terms, the newer the customer is, the more likely they are to leave. From here, it could be interpreted that most customers are vulnerable due to new and weak relationships with the company. For customers to be retained, higher loyalty needs to be established throughout the customer life cycle. Both the T-test ($t = -18.408$, $p<.001$) and logistic regression show that customers who leave pay higher monthly charges, indicating that new customers are price-sensitive and are not receiving the perceived quality associated with the price they are charged.

The chi-square analysis highlights a strong positive association between customers and technical support ($\chi2 = 828.197$, $p < .001$). Without technical care, customers are more likely to leave the company. This shows the importance of service quality and solving issues in a timely manner. For example, if a customer is experiencing an issue with internet service and the company is unable to resolve it in a timely manner, frustration would increase, potentially leading the customer to churn. Similarly, the contract type also has a strong positive association ($\chi2 = 1184.597$, $p < .001$). It is found that customers on a month-to-month contract are more likely to churn than those on longer contracts, such as 1- or 2-year terms. However, there is one observation that needs to be taken into account: are customers on long contracts actually satisfied with the telecommunication company's service, or is it because of the contractual duty? Online Security further confirms that service features play an important role in retention.

Looking through the lens of logistic regression further confirms that these variables influence customer churn. The model explains up to 39% of the variation in churn (Nagelkerke $R^2 = 0.390$), and all variables included are statistically significant. Overall, the model achieved an accuracy of 79.5%, correctly predicting churners and non-churners. However, the model is better at predicting non-churners correctly, 89.3% percent of the time, while churners are only 52.5% of the time. A particularly notable and rather surprising finding is the difference between the chi-square and regression results for online security. After the chi-square analysis was conducted, having online security means the customer is less likely to churn. However, when the same variable was included in a logistic regression, it showed a positive relationship (B = 0.505), indicating that, after accounting for other factors, customers who use online security are more likely to churn. From this, it can be evaluated that customers who opt for more advanced services have higher expectations; if the telecommunication company fails to meet them, the probability of the customer leaving is higher.

Overall, the findings indicate that churn is driven by a combination of early-stage vulnerability, customers' price sensitivity, and gaps in service quality, rather than by a single isolated factor.

Implications:-

1. **Early Customer Retention: -**

New customers haven't yet built a strong relationship with the company, making them more vulnerable to leaving. It is important that the company consider a retention strategy that results in company growth. Since the customers are new, they are still evaluating how the service compares to competitors.

2. **Pricing strategy:-**

Customers who pay more expect higher value. If they don't see the value of their money, they are most likely to leave the company. Hence, it is important that telecommunications can justify its price to new customers. Some ways could include offering them bundles, such as free streaming for 3 months, or making technical support free.

3. **Importance of Technical Support: -**

If customers have access to technical support, churn would decrease. Customer problems not being solved on time or unnecessary delays cause frustration, leading customers to switch to other services. This factor is particularly important for customers on month-to-month contracts, as they have no contractual obligations and can easily switch to another telecommunications provider.

4. **Encourage long-term contracts:**

The company should incentivize customers to move from month-to-month contracts to long-term contracts, as the results clearly showed that those under long-term contracts are less likely to leave. The telecom should offer separate benefits for long-term contract holders, which can then be communicated to those on monthly contracts.

5. **Digital Engagement:**

Encouraging digital features, such as paperless billing, can improve customer engagement, thereby strengthening customer relationships. If the company can get new customers to switch to paperless billing, there is a higher chance of lower churn, as a relationship will slowly form.

6. **Customer Expectations and Perceived Value:**

The positive relationship between churn and online security could indicate that customers expected more than they actually got. If expectations are not met, customers will be frustrated and initially switch to a competitor that provides this service effectively.

Discussion and Recommendations:-

Overall, most of the results were expected and consistent with the identified business problem. Customers who stay longer are less likely to leave, which confirms that loyalty is built over time. Similarly, customers on month-to-month contracts without technical support experienced high churn rates. From this, the company can identify where it is losing customers. Another focus area of telecommunications is online security, as those who have this service are leaving the company.

Churn is not dependent on a single variable; many factors contribute to it. Behavioral, financial, and service-related factors all contribute collectively to a customer's decision to leave. However, the model has its limitations, as it explains only 39% of the variance in churn, leaving the remaining 61% unexplained. These factors were not available in the data set; it could be that the industry in which it operates is highly competitive. Moreover, the model predicts actual churn only 52.5% of the time, suggesting a need for improvement. One variable included in the model is contract, which indicates whether people who stay have a 1-year or a 2-year contract. But it is hard to measure customer satisfaction, as these customers may stay due to contractual obligations. The gap is between customer expectations and the value delivered.

Recommendation

- **Retention Strategy:**

  The analysis shows that customers with shorter tenure (Figure 5) are more likely to leave, highlighting vulnerability in the early stages of employment. One finding from the report is that they are also price-sensitive, and their evaluations are more critical than those of customers with high tenure. As a result, the company should implement targeted onboarding and early engagement strategies, such as personal communication and offering bundles, to strengthen relationships and reduce churn.

- **Pricing Strategy:**

Another finding was that higher monthly charges are strongly associated with churn (as indicated by SPSS analysis), further supported by logistic regression. This means that customers fail to see the perceived value of the service according to the price they are paying. To solve this issue, the company should introduce tiered pricing, with packages tailored to different customers and services, each priced accordingly. This gives customers a choice of what services they want at what price.

- **Technical Support Strategy:**

The analysis shows that customers without technical support are more likely to churn, as shown by both chi-square tests and logistic regression results. These points highlight service quality and issue resolution, which play important roles in customer retention. Therefore, the company should make it easier for customers to reach support and be responsive to the issues they face. This will, in return, decrease their dissatisfaction and churn rate.

References:-

Ahmad, A.K., Jafar, A. and Aljoumaa, K. (2019) 'Customer churn prediction in telecom using machine learning in big data platform,' *Journal of Big Data*, 6(1). https://doi.org/10.1186/s40537-019-0191-6.

King, G.J. *et al.* (2016) 'Dynamic customer acquisition and retention management,' *Production and Operations Management*, 25–25(8), pp. 1332–1343. https://doi.org/10.1111/poms.12559.

Appendix:

Ai Declaration: -

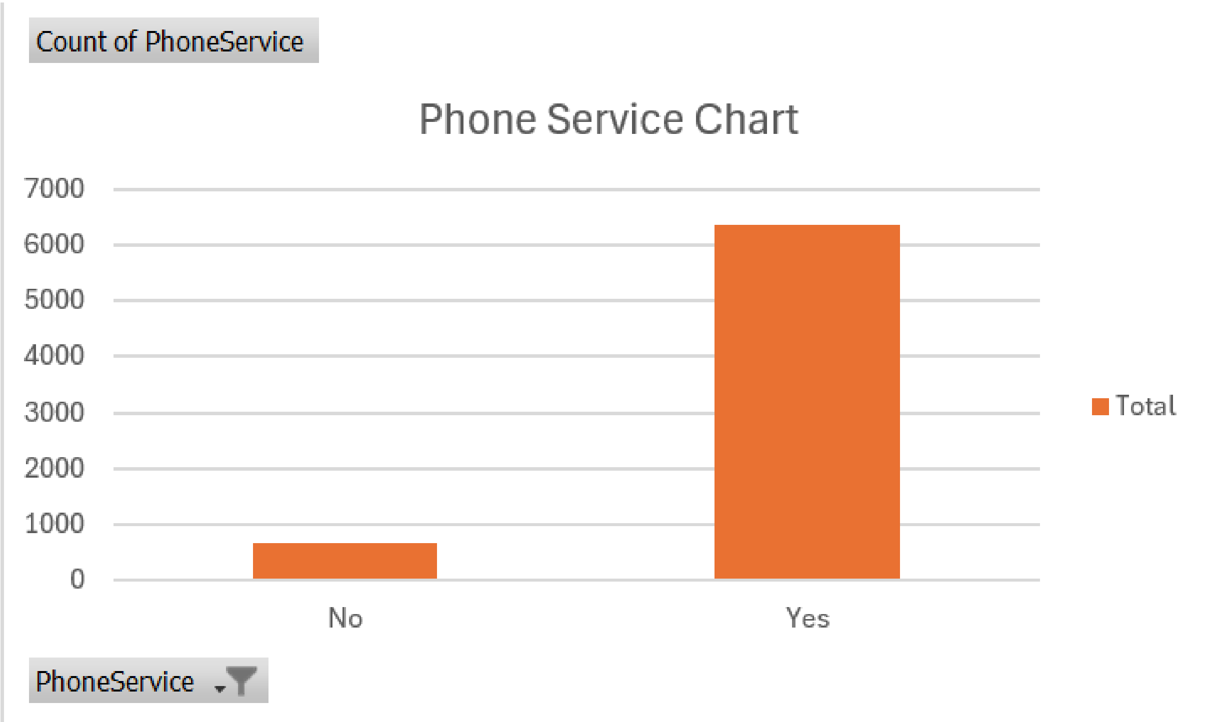
AI was used in this report. It was used to refine my language from a conversational style to a more academic one. All critical points are my own. AI was used to help me understand some of the tables. Statistical software, such as Excel, SPSS, and Tabulate, was used. Other than AI, which helped structure the report and served as a discussion partner.

Tablaue link:-

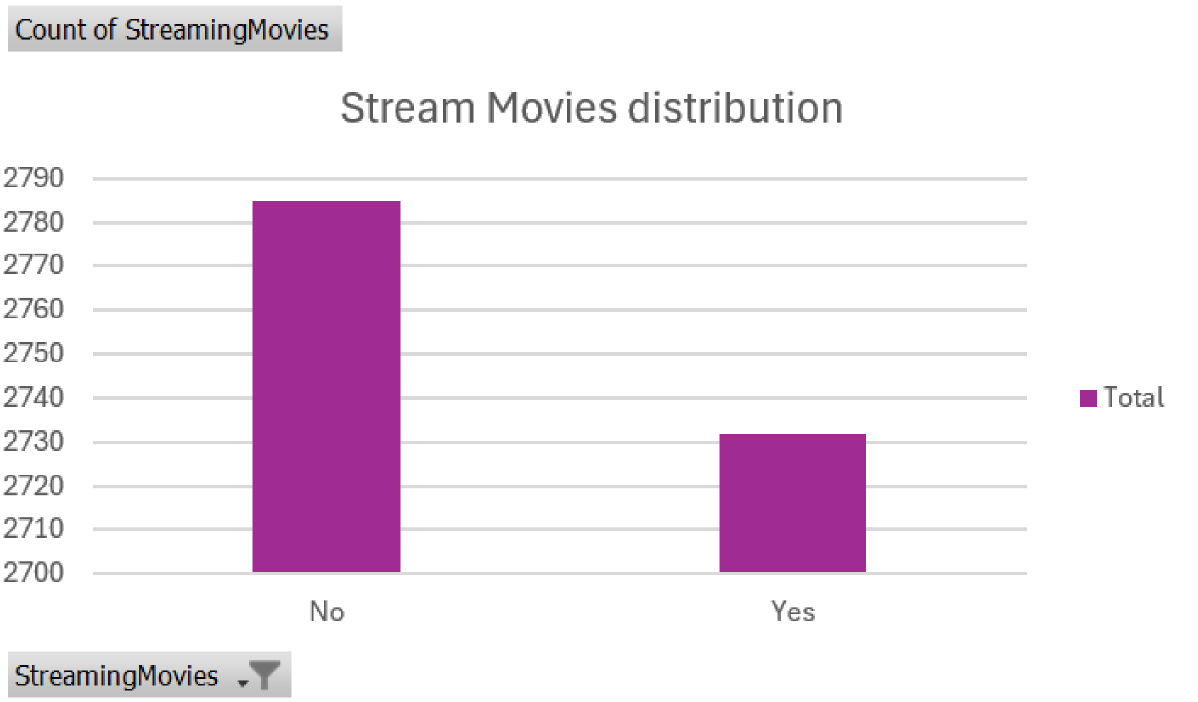
https://public.tableau.com/views/Churndashboard_17767773827610/Dashboard1?:language=en-US&:sid=&:display_count=n&:origin=viz_share_link

Extra graphs

Phone Service



Streaming Movies:



SPSS screenshots for Frequency Tables

gender

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Female	3488	49.5	49.5	49.5
	Male	3555	50.5	50.5	100.0
	Total	7043	100.0	100.0	

SeniorCitizen

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	0	5901	83.8	83.8	83.8
	1	1142	16.2	16.2	100.0
	Total	7043	100.0	100.0	

Partner

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	3641	51.7	51.7	51.7
	Yes	3402	48.3	48.3	100.0
	Total	7043	100.0	100.0	

Dependents

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	4933	70.0	70.0	70.0
	Yes	2110	30.0	30.0	100.0
	Total	7043	100.0	100.0	

PhoneService

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	682	9.7	9.7	9.7
	Yes	6361	90.3	90.3	100.0
	Total	7043	100.0	100.0	

InternetService

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	DSL	2421	34.4	34.4	34.4
	Fiber optic	3096	44.0	44.0	78.3
	No	1526	21.7	21.7	100.0
	Total	7043	100.0	100.0	

PaperlessBilling

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No	2872	40.8	40.8	40.8
	Yes	4171	59.2	59.2	100.0
	Total	7043	100.0	100.0	

Contract

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Month-to-month	3875	55.0	55.0	55.0
	One year	1473	20.9	20.9	75.9
	Two year	1695	24.1	24.1	100.0
	Total	7043	100.0	100.0	